Video Transcript

Video 2: The Dot Product (04:31)

We're going to be manipulating coordinates and we're going to represent things as vectors and sometimes as well as matrices. And so we need to be clear about the direction of the vectors because it's going to become important when we do dot products. Let's take the example of this line. So, Y equals, say, some constant, W0 plus W1 some slope times X. So, that would give us the equation of a line. Let's take a look how we write this in mathematics. So, in mathematics, I always think of vectors as being columns like this. So, when I write Y equals X.w, what I'm thinking of is that I'm thinking of it laid out like this, that X has two columns, one of them will be all ones, and the other column will be x1, x2, x3, and then the ws will also be a column vector. So, I can get the coordinates of Y if you give me the X coordinates like this and w, the two constants. And, I'll just do a dot product. So, my model, if you like, will my predictive model will be a dot product of X and W. But in NumPy, we think of the vectors as being horizontal.

So, we'll have our row of ones and we'll have our row of X coordinates. Now to get the Ys, we need to have W in front. But W as well will be a normal NumPy vector. It'll be a row vector. So, when we do the dot product, we get w0 times 1 plus w1 times x0, and that'll give us y0, and then y1 will be w0 again plus w1 times x1. So, in NumPy we'd write it this way, w.X equals Y or Y equals w.X. It doesn't matter as far as the X which way the equals sign goes, but it does matter notice that here we add X times w and here we have w times X as a dot product.

Now, check your understanding with this little piece of code here. So, here's my w. I'm choosing w0 as one and w1 as two. We're going to have five points. I create a vector, a row vector of ones, and I create an array of zero through four and I stack them. And I put the row of ones on top of the row of Xs.

So, I'm making it look like this and now to get my Ys, I do a dot product of W and X, and we'll print it out. And so here's my X, you'll see it's a two dimensional array. It has a row of ones and the second row is 0, 1, 2, 3, 4. And when we take the dot product of w.X, we'll get Ys as being this 1, 3, 5, 7, 9. So, make sure that you're clear about your vectors. We'll be using NumPy, so I

© MIT xPRO® 2024. All rights reserved.

think you should think of them this way, that they'll be laid out horizontally.

And then when we want to do a predictive model for Y, say, we'll be doing w.X. So, take a look at this piece of code and work through it, change the values. Try it the other way around. Try doing X.w and see what you get. It'll be wrong so be careful. Get this clear in your head and you won't have any trouble. OK, bye for now.

© MIT xPRO® 2024. All rights reserved.